

Foam / prepare and cut

Stock

Use 5 mm FliteBoard Pro paper-faced XPS, 1000 x 600 mm. Measure actual thickness. Do not resize patterns for another thickness.

1 / Print and identify

Print at Actual Size / 100%. Measure the 100 x 20 mm rectangle. Join only matching S-number posters, using row/column labels, paired marks and 10 mm overlap. Follow the index for whole-page patterns.

2 / Make a forming coupon

Keep the outside paper. Peel inside paper in curved areas and gradually relieve or compress the core. Test the tightest intended curve and the adhesive on scrap before cutting the full set.

3 / Cut patterns

Knife-cut CUT lines and transfer REFERENCE marks and panel IDs. Mark OUTSIDE before peeling. DXFs are millimetres with no tool compensation. Use a laser only if the workshop approves this exact XPS and extraction process.

Skin seams

CUT is the finished outside-paper edge: no extra lap or trim, except F23/F24. Those two have a 1 mm paper-only lower lip: remove its foam, keep the paper and trim/fold it against the adjoining lip.

Moving-tail blanks

Cut F31/F32 twice each. Each blank includes 5 mm sacrificial stock around the reference outline. Laminate two 5 mm layers, then trim to the finished outline and shape long edges to R5.

Adhesive

Use XPS-compatible adhesive verified on scrap. Ordinary CA must not contact exposed XPS unless its manufacturer approves that use.

Foam / wings and seams

1 / Dry-fit the frame

Align ribs, spars and both removable wings. Check incidence and symmetry before covering. Match F-series IDs to the assembly map; A/B does not automatically mean left/right.

2 / Form from the leading edge

Keep the outside paper continuous. Peel inside paper only in curved forming zones. Press progressively over the matching frame. Do not stretch a short edge to force a seam.

3 / Relieve the inner core

Shave inner foam where ribs or hinge rails contact it. Preserve outside paper. Fit the whole wing and moving surface together before bonding.

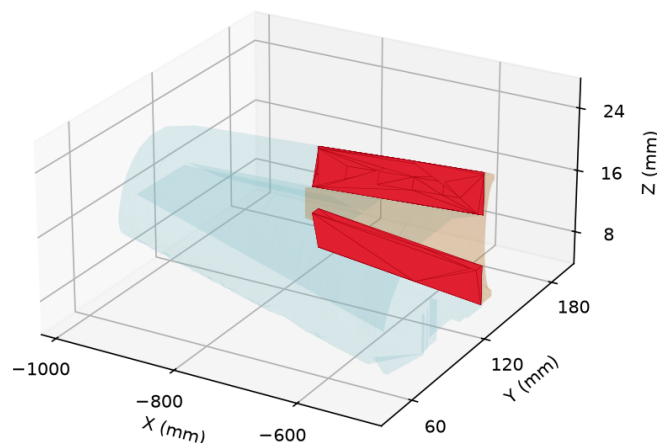
4 / Close seams

Bevel inner foam to the actual meeting angle; there is no universal 45-degree bevel. Butt outside-paper edges. Add separate 15 mm internal paper straps, about 7.5 mm each side. Keep straps out of hinges.

5 / Bond and restrain

Glue progressively with the wing held straight. Peeled board loses stiffness and can warp. Reinforce seams and relief borders without adding material that recreates an interference.

Wing hinge inner core | red = CAD interference to relieve



Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Foam / fuselage and tail

Fuselage order

Fit lower forward, middle and upper panels on the aligned frame. Fit rear-underside bands F27-F30 in numbered order, supporting every inner butt seam. The A42 underside sequence is F19, F20, F33, F21, F22; do not omit F33.

Gear openings

Trace the actual installed gear. Start with 0.5 mm static clearance, then check steering and deflection. Reinforce paper at the opening; foam must not carry gear loads.

Fixed-tail forming tools

Use the removable split collars on a straight jig: T03 at 6.84 mm, T02 at 171 mm and T01 at 335.16 mm along the 342 mm span. Make tools from stiff card or 3 mm scrap plywood; remove both halves after cure.

Fixed-tail skins

Fit the paired skins and root-collar seam. Relieve inner foam so walls meet while preserving the outside shape. Do not add internal ribs into the filled section. Check fit in A12 and symmetry before bonding.

Moving tails

Laminate and shape the two controls. Tape-hinge along the indicated line, fit horns and linkages, and check full travel. Keep glue and seam straps out of moving gaps.

Before closing

Check battery/FC access, cables, joiners, control movement and symmetry. Weigh completed skins and body. Record the physical fit and forming results.

Foam / local fit checks

Wing hinge area

Relieve inner foam around hinge rails. Never cut through outside paper to hide a contact.

Lower centre skin A17

Mark A45/A55 bracket contacts inside and relieve only the foam. Do not sand structural brackets or plywood to fit the skin.

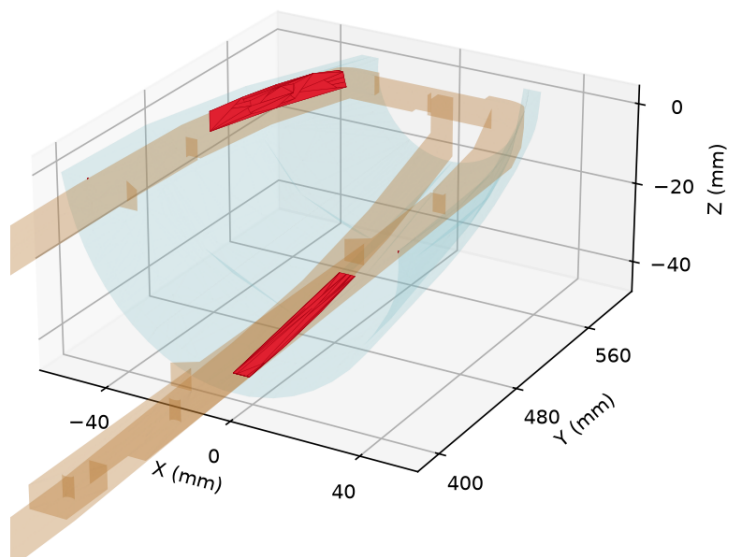
Rear underside A20

Fit A43 and the spine before F27-F30. Transfer contacts into the inner foam. Keep seam support continuous across reliefs.

Final check

Repeat dry-fit with all neighbouring parts installed together. Reinforce relieved-zone edges and seams. Digital pattern development does not establish the stiffness of peeled or relieved foam.

Rear underside inner core | red = CAD interference to relieve



Blue: A20 foam. Tan: A43, A62. Red: trim contact region. View cropped; axes not equal scale.